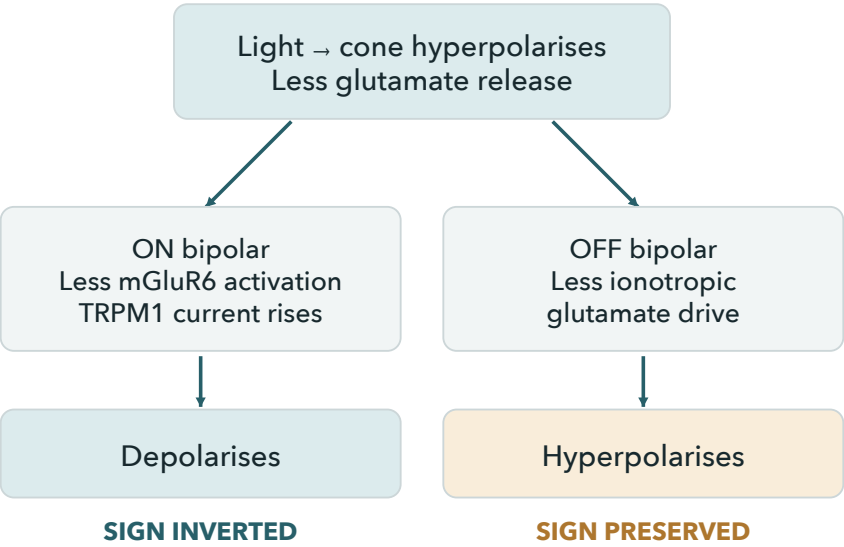


# Retinal contrast circuits

A light increment has opposite signs in the ON and OFF pathways.

## ON and OFF: invert or preserve

### A | One cone signal, two bipolar responses



Photoreceptors and bipolar cells mainly use graded potentials. Most ganglion cells encode output as action potentials.

## Cone bipolar response

| State                 | ON bipolar                                 | OFF bipolar                                         |
|-----------------------|--------------------------------------------|-----------------------------------------------------|
| Dark: more glutamate  | mGluR6 signalling suppresses TRPM1 current | Ionotropic receptors provide depolarising drive     |
| Light: less glutamate | Released from inhibition; depolarises      | Less depolarising drive; hyperpolarises             |
| mGluR6 kept active    | Light-evoked ON response reduced           | Separate OFF receptor pathway can remain responsive |

## Fast check

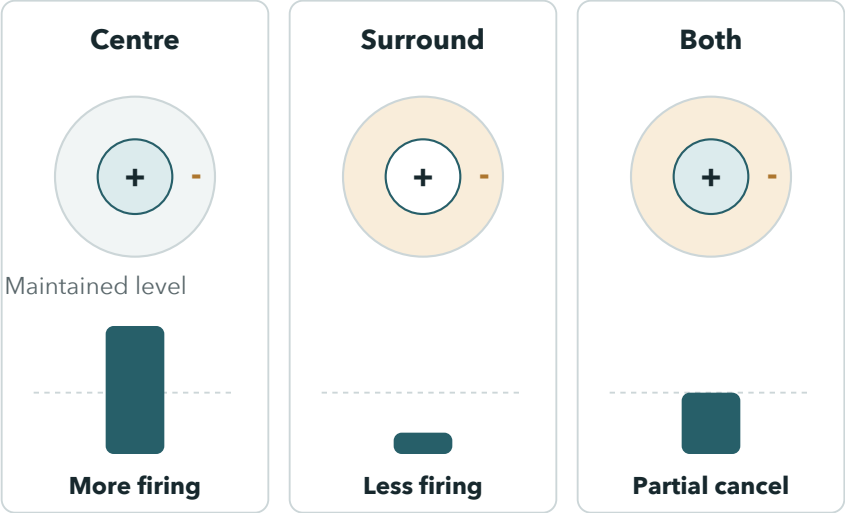
“ON” means preference for a light increment, not that the bipolar cell releases the same response sign as its photoreceptor.

## Figure reference

A: Webvision bipolar pathways; supplied TBL2 retinal processing. Original schematic.

# Centre-surround: encode a difference

## B | ON-centre ganglion responses



Tinted centre or annulus shows the illuminated region. Responses are relative to maintained firing; OFF-centre cells have opposite preference.

## Where is the comparison built?

| Circuit                                 | Main role                                                 |
|-----------------------------------------|-----------------------------------------------------------|
| Horizontal cells: outer plexiform layer | Lateral interactions contribute to antagonistic surrounds |
| Amacrine cells: inner plexiform layer   | Shape timing and other ganglion-cell computations         |
| Ganglion receptive field                | Centre and surround together emphasise local differences  |

## Fast check

A bright centre can increase firing; adding a bright surround can reduce that increase. A large uniform patch often drives less contrast signal than a centre-selective spot.

## Figure reference

B: Webvision ganglion-cell responses. Original stimulus-response schematic.